

Vector Algebra

Question1

Three vectors each of magnitude $3\sqrt{1.5}$ units are acting at a point. If the angle between any two vectors is $\frac{\pi}{3}$, then the magnitude of the resultant vector of the three vector is

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Options:

A.

$9\sqrt{3}$ units

B.

9 units

C.

$\sqrt{6}$ units

D.

3 units

Answer: B

Solution:

Given data

Each vector has magnitude $a = 3\sqrt{1.5} = 3\sqrt{\frac{3}{2}}$.

Angle between any pair of vectors = $\frac{\pi}{3}$.

Let the three vectors be $\vec{A}, \vec{B}, \vec{C}$.

We want the magnitude of $\vec{R} = \vec{A} + \vec{B} + \vec{C}$.

Step 1: Compute $R^2 = |\vec{A} + \vec{B} + \vec{C}|^2$

$$R^2 = A^2 + B^2 + C^2 + 2(\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{C} + \vec{C} \cdot \vec{A})$$

Since all magnitudes are equal:

$$A^2 = B^2 = C^2 = a^2,$$

and

$$\vec{A} \cdot \vec{B} = a^2 \cos\left(\frac{\pi}{3}\right) = a^2 \left(\frac{1}{2}\right).$$

Step 2: Substitute

$$R^2 = 3a^2 + 2 \times 3 \times a^2 \left(\frac{1}{2}\right) = 3a^2 + 3a^2 = 6a^2.$$

Therefore,

$$R = a\sqrt{6}.$$

Step 3: Substitute value of a

$$a = 3\sqrt{1.5} = 3\sqrt{\frac{3}{2}}.$$

So,

$$R = 3\sqrt{\frac{3}{2}} \times \sqrt{6} = 3\sqrt{\frac{3 \times 6}{2}} = 3\sqrt{9} = 3 \times 3 = 9.$$

 **Final Result:**

$R = 9 \text{ units}$

Correct Option: B) 9 units

Question2

A vector perpendicular to the vector $(4\hat{i} - 3\hat{j})$ is

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Options:

A.

$$4\hat{i} + 3\hat{j}$$

B.

$$6\hat{i}$$

C.

$$3\hat{i} - 4\hat{j}$$

D.

$$7\hat{k}$$

Answer: D

Solution:

To find a vector that is perpendicular to $(4\hat{i} - 3\hat{j})$, remember that two vectors are perpendicular if their dot product is zero.

Let's check with $7\hat{k}$. Here, $\hat{k}$ is perpendicular to both $\hat{i}$ and $\hat{j}$.

Compute the dot product: $(4\hat{i} - 3\hat{j}) \cdot 7\hat{k} = 4 \times 0 - 3 \times 0 + 0 \times 7 = 0$.

So, $7\hat{k}$ is perpendicular to $(4\hat{i} - 3\hat{j})$.

Question3

If α, β and γ are the angles made by a vector with x, y and z axes respectively, then $\sin^2 \alpha + \sin^2 \beta =$

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Options:

A.

$$\sin^2 \gamma$$

B.

$$\cos^2 \gamma$$

C.

$$1 + \cos^2 \gamma$$

D.

$$1 + \sin^2 \gamma$$

Answer: C

Solution:

We know that,

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\Rightarrow 1 - \sin^2 \alpha + 1 - \sin^2 \beta + \cos^2 \gamma = 1$$

$$\Rightarrow 1 + \cos^2 \gamma - (\sin^2 \alpha + \sin^2 \beta) = 0$$

$$\Rightarrow 1 + \cos^2 \gamma = \sin^2 \alpha + \sin^2 \beta$$

$$\Rightarrow \sin^2 \alpha + \sin^2 \beta = 1 + \cos^2 \gamma$$

Question4

If the magnitude of a vector **P** is 25 units and its *y*-component is 7 units, then its *x*-component is

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Options:

A.

24 units

B.

18 units

C.

32 units

D.

16 units

Answer: A

Solution:

$$P = |P| = 25 \text{ units}$$

$$P_y = 7$$

$$\text{Since, } \sqrt{P_x^2 + P_y^2} = P$$

$$\Rightarrow P_x^2 + P_y^2 = P^2$$

$$\Rightarrow P_x^2 + 7^2 = 25^2$$

$$\begin{aligned}\Rightarrow P_x &= \sqrt{25^2 - 7^2} \\ &= \sqrt{625 - 49} \\ &= \sqrt{576} = 24 \text{ units}\end{aligned}$$

Question5

The magnitudes of two vectors are A and B ($A > B$). If the maximum resultant magnitude of the two vectors is ' n ' times their minimum resultant magnitude, then $\frac{A}{B} =$

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Options:

A.

$$\frac{n}{n-1}$$

B.

$$\frac{n+1}{n}$$

C.

$$\frac{n^2+1}{n-1}$$

D.

$$\frac{n+1}{n-1}$$

Answer: D

Solution:

We know that,

$$R_{\max} = A + B \text{ and } R_{\min} = A - B$$

$$\text{Since, } R_{\max} = nR_{\min}$$

$$\Rightarrow A + B = n(A - B)$$

$$\Rightarrow \frac{A + B}{A - B} = \frac{n}{1}$$

Applying componendo and dividendo rule,

$$\begin{aligned} \frac{(A + B) + (A - B)}{(A + B) - (A - B)} &= \frac{n + 1}{n - 1} \\ \Rightarrow \frac{2A}{2B} &= \frac{n + 1}{n - 1} \Rightarrow \frac{A}{B} = \frac{n + 1}{n - 1} \end{aligned}$$

Question6

The angle made by the resultant vector of two vectors $2\hat{i} + 3\hat{j} + 4\hat{k}$ and $2\hat{i} + 7\hat{j} - 4\hat{k}$ with X-axis.

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Options:

A. 60°

B. 45°

C. 90°

D. 120°

Answer: B

Solution:

The task is to determine the angle the resultant vector of two given vectors makes with the X-axis.

Given vectors are:

$$\mathbf{A} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\mathbf{B} = 2\hat{i} - 7\hat{j} - 4\hat{k}$$

First, calculate the resultant vector $\mathbf{R}$:

$$\begin{aligned}\mathbf{R} &= \mathbf{A} + \mathbf{B} \\ &= (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}) + (2\hat{\mathbf{i}} - 7\hat{\mathbf{j}} - 4\hat{\mathbf{k}}) \\ &= (2 + 2)\hat{\mathbf{i}} + (3 - 7)\hat{\mathbf{j}} + (4 - 4)\hat{\mathbf{k}} \\ &= 4\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 0\hat{\mathbf{k}} \\ &= 4\hat{\mathbf{i}} - 4\hat{\mathbf{j}}\end{aligned}$$

To find the angle θ between the resultant vector $\mathbf{R}$ and the X-axis, we use the dot product:

$$\cos(\theta) = \frac{\mathbf{R} \cdot \hat{\mathbf{i}}}{|\mathbf{R}| \cdot |\hat{\mathbf{i}}|}$$

Calculate the dot product $\mathbf{R} \cdot \hat{\mathbf{i}}$:

$$\mathbf{R} \cdot \hat{\mathbf{i}} = (4\hat{\mathbf{i}} - 4\hat{\mathbf{j}}) \cdot \hat{\mathbf{i}} = 4$$

Calculate the magnitude of $\mathbf{R}$:

$$|\mathbf{R}| = \sqrt{(4)^2 + (-4)^2} = \sqrt{16 + 16} = \sqrt{32} = 4\sqrt{2}$$

Now, substitute these values into the cosine equation:

$$\begin{aligned}\cos \theta &= \frac{4}{4\sqrt{2}} = \frac{1}{\sqrt{2}} \\ \theta &= \cos^{-1} \left(\frac{1}{\sqrt{2}} \right) \\ \theta &= 45^\circ\end{aligned}$$

Therefore, the resultant vector makes an angle of 45° with the X-axis.

Question7

If $|\mathbf{P} + \mathbf{Q}| = |\mathbf{P}| = |\mathbf{Q}|$ then the angle between $\mathbf{P}$ and $\mathbf{Q}$ is

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Options:

A. 0°

B. 120°

C. 60°

D. 90°

Answer: B

Solution:

Given that $|\mathbf{P} + \mathbf{Q}| = |\mathbf{P}| = |\mathbf{Q}|$, let's analyze the problem step-by-step:

Begin with the equation:

$$|\mathbf{P} + \mathbf{Q}|^2 = |\mathbf{P}|^2$$

Expanding $(\mathbf{P} + \mathbf{Q}) \cdot (\mathbf{P} + \mathbf{Q})$:

$$(\mathbf{P} + \mathbf{Q}) \cdot (\mathbf{P} + \mathbf{Q}) = \mathbf{P} \cdot \mathbf{P} + 2\mathbf{P} \cdot \mathbf{Q} + \mathbf{Q} \cdot \mathbf{Q} = P^2 + 2\mathbf{P} \cdot \mathbf{Q} + Q^2$$

Since $|\mathbf{P}| = |\mathbf{Q}|$, let them both be P (i.e., $Q = P$). Thus, the equation becomes:

$$P^2 + 2\mathbf{P} \cdot \mathbf{Q} + P^2 = P^2$$

Simplify and rearrange:

$$2\mathbf{P} \cdot \mathbf{Q} + P^2 = 0$$

Substitute $\mathbf{P} \cdot \mathbf{Q} = PQ \cos \theta$:

$$2PQ \cos \theta + P^2 = 0$$

Solve for $\cos \theta$:

$$2P^2 \cos \theta = -P^2$$

$$\cos \theta = -\frac{1}{2}$$

Recognize that $\cos \theta = -\frac{1}{2}$ corresponds to:

$$\theta = 120^\circ$$

Thus, the angle between $\mathbf{P}$ and $\mathbf{Q}$ is 120° .

Question8

The component of a vector $\mathbf{P} = 3\hat{i} + 4\hat{j}$ along the direction $(\hat{i} + 2\hat{j})$ is

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Options:

A. $\frac{8}{\sqrt{5}}$

B. $\frac{11}{\sqrt{5}}$

C. $\frac{11}{2}$

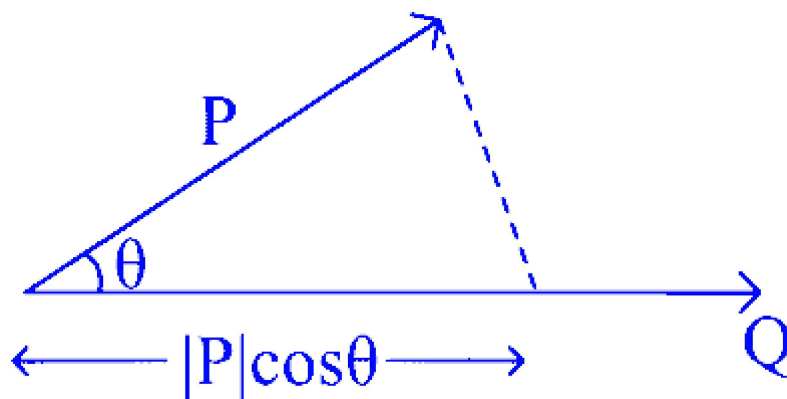
D. $\sqrt{10}$

Answer: B

Solution:

Given, $\mathbf{P} = 3\hat{i} + 4\hat{j}$

Let $\mathbf{Q} = \hat{i} + 2\hat{j}$



The component of $\mathbf{P}$ along the $\mathbf{Q}$ is given as $|\mathbf{P}| \cos \theta$.

Hence, we know that,

Hence, $\mathbf{P} \cdot \mathbf{Q} = |\mathbf{P}||\mathbf{Q}| \cos \theta$

$$\begin{aligned} \Rightarrow |\mathbf{P}| \cos \theta &= \frac{\mathbf{P} \cdot \mathbf{Q}}{|\mathbf{Q}|} \\ &= \frac{(3\hat{i} + 4\hat{j}) \cdot (\hat{i} + 2\hat{j})}{\sqrt{1^2 + 2^2}} = \frac{3 + 8}{\sqrt{5}} = \frac{11}{\sqrt{5}} \end{aligned}$$

Question9

If two vectors $\mathbf{A}$ and $\mathbf{B}$ are mutually perpendicular, then the component of $\mathbf{A} - \mathbf{B}$ along the direction of $\mathbf{A} + \mathbf{B}$ is

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Options:

A. $\sqrt{|A|^2 + |B|^2}$

B. $\sqrt{|A|^2 - |B|^2}$

C. $\frac{|A|^2 - |B|^2}{\sqrt{|A|^2 + |B|^2}}$

D. $\frac{|A|^2 + |B|^2}{\sqrt{|A|^2 - |B|^2}}$

Answer: C

Solution:

The two vectors **A** and **B** are mutually perpendicular to each other

$$\mathbf{A} \cdot \mathbf{B} = 0$$

$$\Rightarrow AB \cos \theta = 0$$

$$\Rightarrow \cos \theta = 0 \Rightarrow \theta = 90^\circ$$

$$\begin{aligned} |\mathbf{A} - \mathbf{B}| &= \sqrt{A^2 + B^2 - 2AB \cos 90^\circ} \\ &= \sqrt{A^2 + B^2} \end{aligned}$$

Unit vector along **A - B**

$$= \frac{\mathbf{A} - \mathbf{B}}{|\mathbf{A} - \mathbf{B}|} = \frac{\mathbf{A} - \mathbf{B}}{\sqrt{A^2 + B^2}}$$

component of **(A + B)** in direction of **(A - B)** will be given as,

$$\begin{aligned} &= \frac{(\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B})}{|\mathbf{A} - \mathbf{B}|} \\ &= \frac{\mathbf{A} \cdot \mathbf{A} - \mathbf{A} \cdot \mathbf{B} + \mathbf{B} \cdot \mathbf{A} - \mathbf{B} \cdot \mathbf{B}}{\sqrt{A^2 + B^2}} \\ &= \frac{A^2 - B^2}{\sqrt{A^2 + B^2}} = \frac{|\mathbf{A}|^2 - |\mathbf{B}|^2}{\sqrt{|\mathbf{A}|^2 + |\mathbf{B}|^2}} \end{aligned}$$

Question10

Which of the following is not true about vectors **A**, **B** and **C** ?

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Options:

A. $(\mathbf{A} \cdot \mathbf{A})(\mathbf{B} \cdot \mathbf{C})$ is a scalar value.

B. $(\mathbf{A} \times \mathbf{B}), (\mathbf{B} \times \mathbf{C})$ is a scalar value.

C. $(\mathbf{A} \times \mathbf{C}) \times (\mathbf{B} \times \mathbf{C})$ is a scalar value.

D. $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ is a vector value.

Answer: C

Solution:

In option (a)

$(\mathbf{A} \cdot \mathbf{A})(\mathbf{B} \cdot \mathbf{C})$

$(A^2)(BC \cos \theta)$

$= A^2 BC \cos \theta$ Scalar value

In option (b)

$$\begin{aligned}(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{B} \times \mathbf{C}) &= (AB \sin \theta \hat{\mathbf{n}}) \cdot (BC \sin \phi \hat{\mathbf{k}}) \\ &= (AB \sin \theta)(BC \sin \phi)(\hat{\mathbf{n}} \cdot \hat{\mathbf{k}})\end{aligned}$$

where, $\hat{\mathbf{n}}$ is unit vector perpendicular to both vector $\mathbf{A}$ and $\mathbf{B}$ and $\hat{\mathbf{k}}$ is unit vector perpendicular to both vector $\mathbf{B}$ and $\mathbf{C}$.

$\therefore \hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = \text{scalar value}$

$\therefore (\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{B} \times \mathbf{C})$ is a scalar value

$$\begin{aligned}(\mathbf{A} \times \mathbf{C}) \times (\mathbf{B} \times \mathbf{C}) &= (AC \sin \theta_1) \hat{\mathbf{r}}_1 \times (BC \sin \theta_2) \hat{\mathbf{r}}_2 \\ &= (AC \sin \theta_1)(BC \sin \theta_2) (\hat{\mathbf{r}}_1 \times \hat{\mathbf{r}}_2)\end{aligned}$$

$\hat{\mathbf{r}}_1 \times \hat{\mathbf{r}}_2$ is a vector product, hence it gives a vector value.

Therefore, $\mathbf{A} \times \mathbf{C} \times (\mathbf{B} \times \mathbf{C})$ is a vector value.

In option (d) $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ is vector triple product which gives a vector value.

Question 11

One of the rectangular components of a force of 40 N is $20\sqrt{3}$ N. What is the other rectangular component?

Options:

- A. 10 N
- B. 20 N
- C. 30 N
- D. 25 N

Answer: B

Solution:

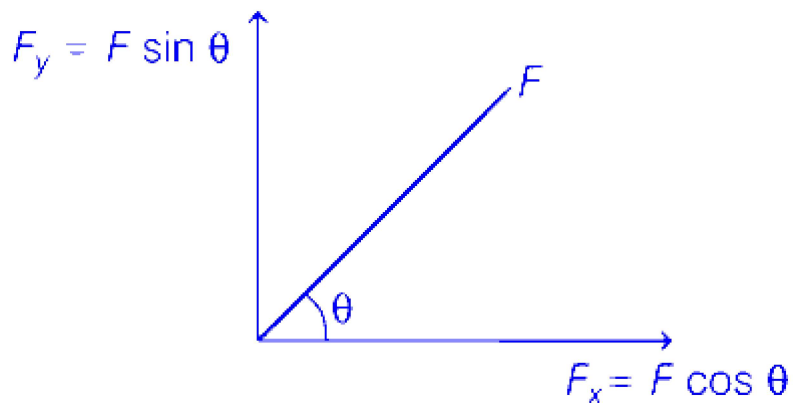
Given, applied force, $F = 40 \text{ N}$

Horizontal component of force, $F_x = 20\sqrt{3}$

and θ be the angle of force from horizontal X-axis.

According to given diagram,

The component of F along X-axis will be



$$F_x = F \cos \theta$$

$$\Rightarrow 20\sqrt{3} = 40 \cos \theta \Rightarrow \sqrt{3}/2 = \cos \theta$$

$$\therefore \theta = \cos^{-1}(\sqrt{3}/2) = 30^\circ$$

And the component of F along Y-axis

$$F_y = F \sin \theta$$

$$= F \sin 30^\circ$$

$$= 40 \times 1/2 = 20 \text{ N}$$
